

What a ZIP Code Is, and What It Is Not

A routing key for mail trucks, and everything it gets asked to stand for

84-355 Democracy's Data

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A letter addressed to 1600 Pennsylvania Avenue NW, Washington, DC 20500 will reach the White House. That is what the five digits are for, and they do it well.

Now ask a different question of the same five digits: **how many people live in 20500?**

The question sounds ordinary. It is the kind of question a survey asks, an insurer prices, a public health study controls for, and a redistricting proposal would build a congressional district out of. It has no answer. Not an unknown answer — no answer, because 20500 is not a piece of ground. It is a routing instruction that happens to be written as a number, and there is nothing underneath it to count.

This chapter is about the distance between those two questions, and about how much of American data analysis is quietly standing on the wrong side of it.

01 Where the file comes from, and what it is for

The list of ZIP codes counted in these pages does not come from the Postal Service, because the Postal Service does not give its list away. The authoritative register is a product called the *City State Product*, and USPS sells it on subscription. What stands in for it here, as in most open work, is GeoNames — a gazetteer, which is to say a dictionary of places. It is built on the open-content model: compiled from public sources, corrected by volunteers, and given away whole. Everything else in this chapter is the Census Bureau's work, and where the Bureau's own files come from is a chapter of its own.

Who is in it, and how they got there. Codes, not people. A ZIP code is in this list because it was seen in use: GeoNames compiles its American postal file from address data, not from the Postal Service's register. A code enters by appearing on addresses and leaves by no longer appearing on them, which is not the same thing as being issued or retired.

Who it was made for. Nobody commissioned it. It exists because a great deal of ordinary work needs a national postal list and cannot pay a subscription for one: geocoding, mapping, matching a survey answer to a place. It has become the list that open research quietly runs on.

What it costs to get. Nothing. One download, no key, no account, under a Creative Commons licence whose single condition is attribution. The register it stands in for is priced; the stand-in is free; and that difference, more than any measure of quality, decides which list most analyses actually use.

What has already been done to it. It is a compilation, not the register. Nobody at GeoNames assigns a ZIP code or retires one, and nobody there can say that a code does not

exist — only that it has not been seen. Two compilations built from different address data will disagree about the tail of the list, so the count of codes below is a good likeness of the register, not a copy of it. What that substitution costs, and what it says that anyone had to make it, is a question the end of this chapter comes back to.

02 Two lists, and only one of them is a map

Two national lists exist. They are not the same list.

List	What it is	Count
postal codes in circulation	routing keys, compiled from mail addresses	41,488
ZIP Code Tabulation Areas	areas, built from census blocks	33,791
in both	a code that is also an area	33,642
postal code with no tabulation area	a code that names no ground at all	7,846
tabulation area with no postal code	ground the compilation does not list	149

The first is a list of **routing keys**. The Postal Service assigns a ZIP code to a set of delivery stops. That might be a carrier’s walking route, a rural route, a bank of post office boxes, or a single building that generates enough mail to justify its own. A code is a label on a bundle of addresses, and it is revised whenever revising it moves mail more efficiently.

The second is a list of **areas**. ZIP Code Tabulation Areas are the Census Bureau’s approximation of the first list. For each census block, take the ZIP code appearing on the most addresses inside it, then glue together the blocks that share an answer. A ZCTA is therefore a real shape with real boundaries. But they are the Bureau’s boundaries, worked out from where the mail goes. The Postal Service’s boundaries do not exist as boundaries at all.

Almost every map of ZIP codes you have ever seen is the second list. Almost every claim made about ZIP codes is about the first.

Question	ZIP code	ZCTA
Who assigns it?	U.S. Postal Service	U.S. Census Bureau
What is it attached to?	delivery stops	census blocks
When does it change?	whenever routing changes	each decennial census
Is there a boundary?	no	yes, and it is drawn
Is there a population?	no — nobody counts delivery stops	yes — the census counts the blocks it is made of

That last row is the one that keeps the confusion alive, because the answer is yes. **The 2020 census is tabulated by ZCTA**, and 334,726,586 people were counted inside one. Every population figure below is that count. So a ZIP area does have a population, published by the Census Bureau, and it is a real count rather than a model.

What it is a count of is the catch. It is the population of **the blocks the Bureau assigned to that code** — the tabulation area, not the ZIP code, which has no area to count anybody in. And a second federal source answers the same question differently.

The American Community Survey publishes an estimate for the same 31,972 areas three years later. The two readings differ by a median of 6.0%, and 11,448 of them differ by more than a tenth. Neither is wrong. They are three years apart and one of them is a sample. But “the population of ZIP code 15217” is a phrase with at least two published answers, and nothing in either file says which one a study used.

03 The first digit is a map of the mail

Read a ZIP code left to right and it is a set of instructions for narrowing down, exactly like a postal address read from the bottom up.

- **The first digit** picks one of ten national areas.
- **The first three digits** pick a sectional center facility — the plant that sorts for a region. There are 896 of them among the tabulation areas.
- **The last two** pick the delivery unit inside that region.

Which states sit under which leading digit is not a convention anybody has to remember. It is in the data, and the figure below is the first digit of every sectional prefix drawn where its mail is.

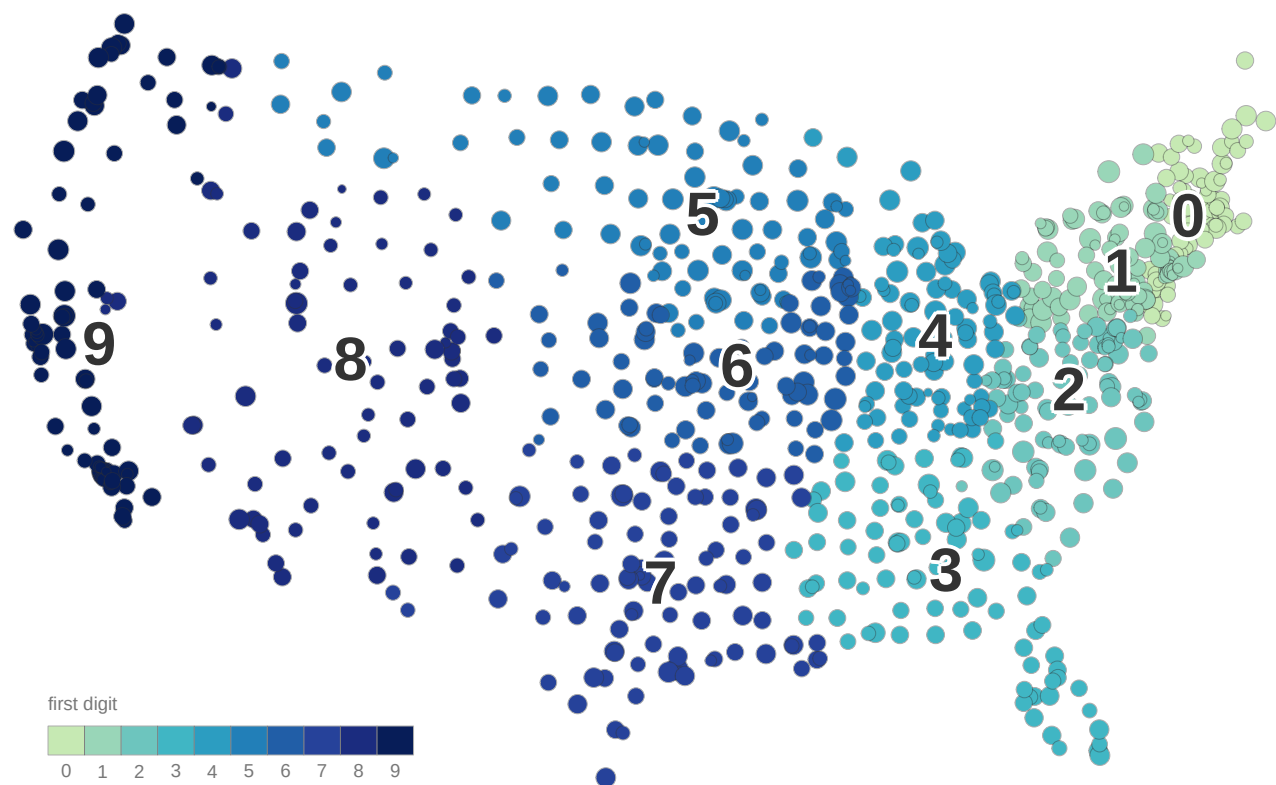


Figure 1. The mail system, drawn. One mark per three-digit prefix, 884 of them, placed at the population-weighted centre of the areas that carry it and sized by how many areas that is. Colour is the leading digit. 12 prefixes are outside the frame, in AK, HI, PR. They are left undrawn rather than moved somewhere they are not.

Nobody drew that map, and it is a real map. Ten bands, running west from the north-east corner, each one a coherent block of territory. It comes from nothing but the require-

ment that mail be sorted in stages. A bag routed to area 1 can be sorted again into 15, then into 152, then into 15213, and each stage only has to look at one more digit.

First digit	ZIP areas	Sorting prefixes	States with at least 1% of them
0	2,583	83	NJ MA ME CT VT NH PR RI
1	3,726	99	PA NY DE
2	3,452	98	VA NC WV MD SC DC
3	3,483	91	FL GA AL TN MS
4	3,812	96	OH MI IN KY
5	3,766	86	IA MN WI ND SD MT
6	3,721	86	IL MO KS NE
7	3,808	93	TX OK AR LA
8	2,252	75	CO AZ NM UT ID NV WY
9	3,188	89	CA WA OR AK HI

So the reel is right, and it is worth being precise about why it is right. The map is real because the mail system is real. **What it is a map of is mail.** It is not a map of communities, or of markets, or of anything drawn with a person in mind. Every use of a ZIP code as though it were one of those things borrows the authority of this figure for a claim the figure does not make.

04 The codes that name no ground

Look again at the two lists. There were more routing keys than areas, and the difference is not rounding.

Before you read on, commit to a number. There are 41,488 ZIP codes in circulation. Some of them cannot be drawn as areas at all, because the addresses they serve do not cover any territory: a single office tower, a wall of post office boxes, a mail channel that follows the Navy. **What share of the 41,488 do you think those are?** One in a thousand? One in a hundred? Write a percentage down before you turn the page.

ZIP code	Listed as	What it actually is
20500	Washington, DC	the White House
10118	New York, NY	the Empire State Building
12345	Schenectady, NY	General Electric, Schenectady
20898	Gaithersburg, MD	boxes rented at one Maryland post office
09096	APO AE	Army and Air Force post office, Europe

7,846 of the 41,488 name no ground, or 18.9%. Close to one in five. The Census Bureau does not fail to draw them; there is nothing to draw. A code assigned to the Empire State

Building covers the footprint of one building and the hundred businesses stacked inside it. A code assigned to a bank of post office boxes covers the boxes, whose renters live wherever they live.

509 of them are military and diplomatic mail. 09096 is listed at longitude 8.249, latitude 50.083, which is in Germany. It is an American ZIP code, and there is no American ground under it.

These are not exotic edge cases parked in a corner of the country. 52 states have at least one, and the state with the most is CA, with 791.

The reverse case exists too and is smaller: 149 tabulation areas carry a code the postal compilation no longer lists. Every one of them was a live ZIP code when the areas were drawn, and has left circulation since. The next section explains why that is guaranteed.

What this does to an analysis is specific and easy to miss. Join a table of ZIP codes to a table of ZIP areas and roughly one row in five silently fails to match — and the rows that fail are not a random sample. They are office buildings, PO box clusters, and military addresses, which is to say they are businesses, people without stable residential delivery, and service members. A join that drops them is a join that has quietly excluded three specific populations and reported nothing.

05 How a ZIP code becomes a ZCTA

The two words are used interchangeably, including by people who know better, and almost every map labelled “ZIP codes” is really the second of them. They are different objects made by different agencies for different reasons, and the translation between them is a specific procedure with specific losses.

Here is the whole of the difference in one table.

	ZIP code	ZCTA
What is it?	a label on a set of delivery addresses	an area made of whole census blocks
Who defines it?	U.S. Postal Service	U.S. Census Bureau
How does it exist in a file?	a field on each address point	a polygon
How many are there?	41,488	33,791
How many have no counterpart?	7,846 have no tabulation area	149 carry a code no longer in circulation
Where does its edge fall?	nowhere – it has no edge	on a census block boundary, always
When does it change?	whenever routing changes	once, at each decennial census
Does it cover the country?	only where mail is delivered	inhabited land; empty pieces over 2 sq mi are left out
Does it have a population?	no – nobody counts delivery stops	yes – the census count of its blocks

The procedure has four steps, and the Bureau states each of them.

1. **A ZIP code enters the Bureau's files as an attribute of an address**, not as a shape.
Every delivery point in the Master Address File carries the ZIP code the Postal Service

uses for it. At this stage the data is a scatter of points with labels, and nothing about it is an area.

2. **Each census block takes the ZIP code carried by the most addresses inside it.** This is where the losses happen, and they are unavoidable: a block whose addresses are split between two ZIP codes goes entirely to one of them, and the minority is gone. A ZIP code that never wins a block anywhere — a single office tower, a bank of post office boxes — gets no ground at all, which is the 7,846 from the previous section.
3. **Blocks with the same answer are glued together** and the result takes those five digits. Because the pieces are whole blocks, **a ZCTA boundary is always a census block boundary** — while a ZIP code's notional edge, if it had one, would run down the middle of streets, because that is how a carrier's route is built.
4. **Land nobody lives on can be left out.** All inhabited ground gets covered, but empty stretches over two square miles are potentially left unassigned, which is why the tabulation areas do not tile the country.

Only codes valid on 1 January 2020 were used, which is why 149 tabulation areas now carry a code that has since left circulation.

Step 3 is the one worth seeing, because it is what makes a ZCTA an object you can measure and a ZIP code not.



Figure 2. The pieces a ZIP area is made of. The 301 census blocks assigned to 15213 and the 6 assigned to 15260, with the published outline of each ZIP area drawn over them. The outline never crosses a block; it runs along block edges, because it was made by gluing blocks together.

The claim can be checked rather than admired. Adding up the population of those 301 blocks gives 31,075 people, which is exactly the published population of 15213, and the 6 blocks of 15260 give exactly 29. It is the same people counted twice, once by block and once by ZIP area, and the two counts agree to the person.

Commercial ZIP boundary files exist too, and they are inferences of a different kind — usually drawn from address point files or delivery route geometry, often proprietary, and generally not identical to the Bureau's. Two ZIP code maps of the same city need not agree, and neither is wrong, because there is no original they are both copying.

There is a consequence the gluing cannot avoid, and it is visible on the ground.

ZIP	What is in it	Sq km	Counted in 2020	Pieces
15213	Oakland: Carnegie Mellon, most of Pitt's campus	6.32	31,075	1 outline with 2 holes
15260	University of Pittsburgh mail	0.24	29	2 disconnected pieces

06 15213, and the two holes in it

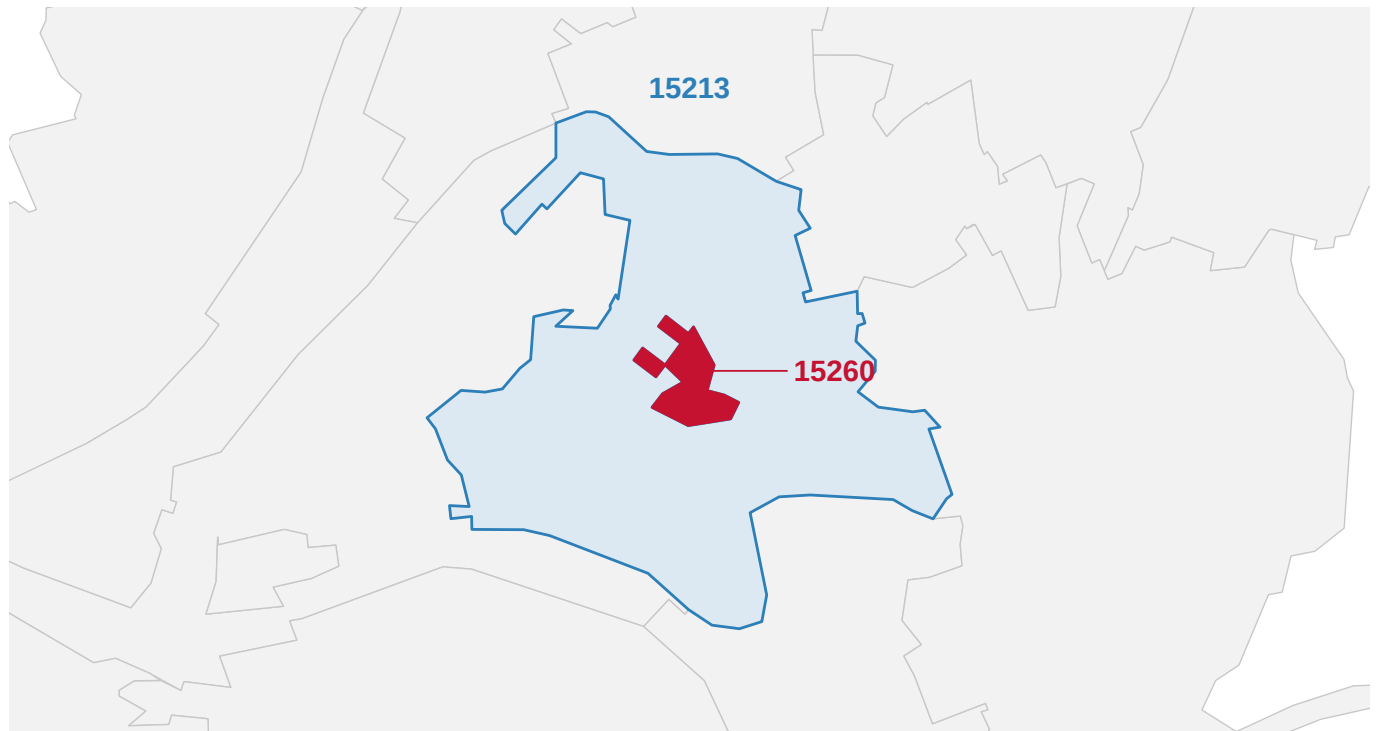


Figure 3. A ZIP code with two holes in it. ZIP area 15213 in blue, the ZIP area 15260 in red, neighbouring areas in gray. 15213 is a single shape with 3 rings: one outline and two holes. The two holes are 15260, which is 2 disconnected pieces and touches nothing else.

This is where this course is taught. 15213 is Oakland: Carnegie Mellon, the museums, most of the University of Pittsburgh's ground. 15260 is the University of Pittsburgh's *mail*: a code assigned to the university's own delivery. Its tabulation area comes out as two islands of campus buildings, punched out of the middle of the neighbourhood around them. The 2020 census counted 29 people in it, against 31,075 in the ZIP area whose middle it occupies.

Nothing here is a mistake. Every step of it is the correct application of a sensible rule. It is simply what happens when you ask "where is this delivery route?" and require the answer to be a polygon.

Now imagine using these two shapes as building blocks for anything. A district drawn to include 15213 has a hole in it unless it also takes 15260. A statistic reported for 15213 describes a neighbourhood with its university buildings cut out. A survey respondent who writes "15213" and a survey respondent who writes "15260" may work in the same building.

07 It does not fit a county, and sometimes not a state

The census geography of the United States nests: a block is inside a block group, inside a tract, inside a county, inside a state, and each identifier contains its parents. ZIP areas were drawn against delivery routes, which respect none of that.

Counted above	In 2+ counties	% of ZIP areas	In 2+ states	% of people
0%	10,180	30.1	137	25.6
1%	9,441	27.9	117	22.1
5%	7,675	22.7	89	16.2

10,180 of the 33,791 ZIP areas have land in more than one county, or 30.1% of them. And 25.6% of everybody who lives inside a ZIP area at all lives in one of them. 137 cross a state line. The widest reaches into 6 counties.

The threshold column is there because it has to be. A relationship file records every overlap, including the few square metres where two boundaries follow the same riverbank slightly differently, so a count with no threshold counts digitising noise as a crossing. Raising the bar to five per cent of the ZIP area's own land removes about a quarter of the crossings and leaves 7,675. The conclusion is not sensitive to that choice. The number is.

There is a stranger fact underneath. Tabulation areas do not cover the country.

What	Value
counties holding land under no ZIP area	729
square kilometres under no ZIP area	1,845,990
as a share of the country's land	20.2%
of that land, the share in pieces over two square miles	99.99%

A fifth of the United States has no ZIP code over it. This follows from step four. Inhabited ground is always covered; empty stretches larger than two square miles are the ones that may be left out — and the last row is that rule caught in the act. 99.99% of the uncovered land lies in pieces above the Bureau's own stated threshold, which is about as close as a documentation page and a data file ever come to agreeing.

So the fifth is not a gap in the record. It is national forest, range, desert and tundra, declared out of scope by a rule written down in advance. But if you colour a map of the country by ZIP code, that fifth is a hole. The usual fix is to snap it to the nearest ZIP area, which invents coverage the source is explicitly declining to claim.

08 It does not stay where you left it

A census tract is redrawn every ten years, and everybody knows it. A ZIP code changes whenever routing changes, and it is nobody's job to tell you.

What	Count
codes with an area in 2010	33,144
codes with an area in 2020	33,791
2010 codes gone by 2020	204
2020 codes that did not exist in 2010	851
codes in both, whose area is unchanged	2,028
codes in both, that kept less than 95% of their 2010 land	14,960

Between the 2010 and 2020 vintages, 204 codes stopped having an area and 851 acquired one for the first time. That is the small part. The large part is that of the codes present in both, **only 2,028 came through the decade with the same shape**. The median survivor holds 95.7% of the land it used to hold; 14,960 kept less than nineteen twentieths of it.

So “15217” in a 2011 file and “15217” in a 2021 file are the same string and not the same place, and nothing in either file says so. A time series by ZIP code is a time series over a moving denominator — a bottom number that will not hold still, and the movement is invisible because the label did not change.

09 It does not fit a city

Grofman and Cervas (2021) argue against a proposal to build congressional districts out of ZIP codes, and report a small piece of fieldwork on their own city. Pieces of seventeen ZIP codes are in Irvine, California, and by their best judgment only two are wholly inside it. The claim is recomputed below against the Bureau’s own relationship file — which measures tabulation areas, not ZIP codes, and so is expected to differ a little.

City	Counted above	ZIP areas with land inside	Of those, wholly inside
Irvine	0%	15	3
Irvine	1%	11	3
Irvine	5%	10	3
Pittsburgh	0%	33	13
Pittsburgh	1%	32	13
Pittsburgh	5%	30	13

11 ZIP areas have a real piece inside Irvine and 3 are wholly within it. The published figures were seventeen and two. **The gap between the two accounts is not an error in either of them.** The paper counted ZIP codes as the Postal Service describes them, and this counts the Bureau’s approximation of the same thing. The two lists were never the same list. That the answers land in the same neighbourhood is the reassurance available; that they differ at all is the chapter.

For Pittsburgh: 32 ZIP areas have a real piece inside the city limits, and 13 are wholly inside.

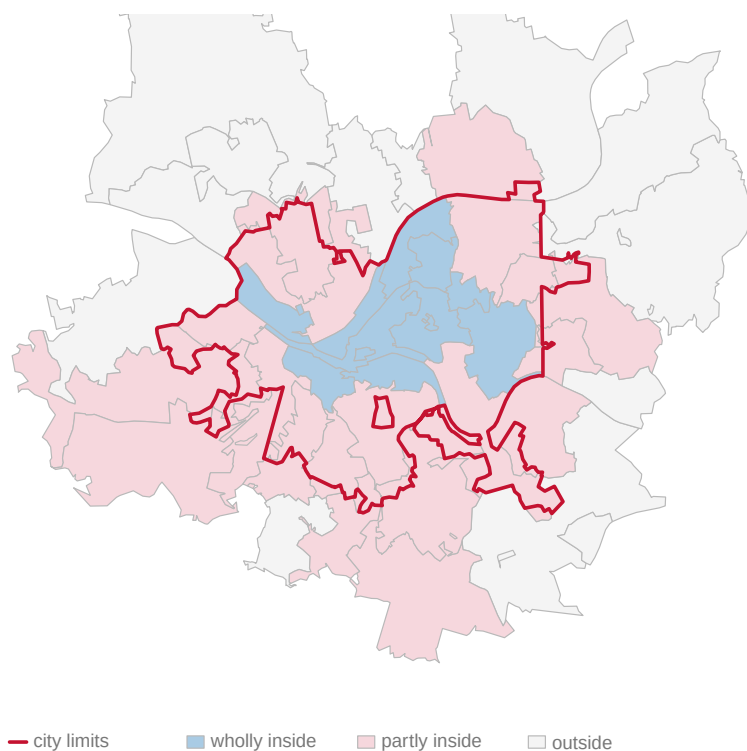


Figure 4. The City of Pittsburgh, and the ZIP areas that do not fit it. The city limits in red; every ZIP area near them shaded by whether it is wholly inside, partly inside, or outside. 32 have a real piece inside the line and 13 are wholly within it.

Neither city is unusual. Across the hundred largest cities, the median has 20 ZIP areas with a real piece inside it and 38.4% of them wholly inside; 4 of the hundred contain no ZIP area whole at all.

10 It does not fit a district

Which brings the argument to the case this book cares about. Curiel and Steelman (2018, 2020) propose building congressional districts out of ZIP codes, on the ground that ZIP codes are how constituents and representatives find each other, and that splitting them damages the link. Grofman and Cervas (2021) reply that the evidence does not support the claim and that the unit is unsuited to the job.

The measurement below is the one thing in this chapter with no relationship file behind it. The Bureau publishes none between ZIP areas and congressional districts. So it is done by overlapping the shapes directly, and the threshold is reported rather than buried.

Counted above	PA ZIP areas	Split	% of areas	% of people	Most districts
1%	1,833	236	12.9	23.4	3
2%	1,833	223	12.2	22.6	3
5%	1,833	198	10.8	19.2	3

223 of Pennsylvania's 1,833 ZIP areas are split between two or more of the state's 17 congressional districts, or 12.2% of them. But those areas hold 22.6% of the people. The

split ZIP areas are the populous ones, which is what you would expect. A district boundary has to run somewhere, and where it runs through a city it runs through the ZIP areas with the most people in them.

Take the argument seriously in both directions and it points the same way.

If splitting ZIP codes damages the link between a voter and their representative, then under the current map that damage falls on roughly a fifth of Pennsylvanians. It is concentrated in exactly the dense places where a district line has no choice but to cut something.

If you fix it by drawing districts from whole ZIP codes, you buy that with everything above. Units that cross county lines, 30.1% of them, and occasionally state lines. Units that leave a fifth of the state's land uncovered. Units with holes in them, and units in two pieces.

Units whose boundaries move between censuses with no announcement. And roughly one code in five with no area to use at all. Most state constitutions require line-drawers to keep counties and cities whole where they can. ZIP areas do not nest inside either.

And there is the question Grofman and Cervas put most directly, which no amount of data can settle for you: **do you know the boundary of your own ZIP code?** Not the code — the boundary. Which streets are in it, where it stops. The argument for ZIP-code districting rests on voters recognising the unit. The authors, who study redistricting for a living, report that before writing the paper neither of them could have drawn the edges of the ZIP code they live in.

11 Why it is used anyway

None of this is obscure, and ZIP codes remain the most-used geography in American applied work. The reason is not ignorance. It is that **the ZIP code is the only piece of geography an ordinary person knows about themselves**.

Ask someone for their census tract and you will get nothing. Ask for their county and about half of urban respondents will guess wrong. Ask for their ZIP code and you get five digits, instantly, correctly, from nearly everybody. It is the cheapest geographic question in survey research, and the answer arrives pre-coded.

So a survey collects ZIP codes because that is what people can supply. An insurer rates by ZIP code because that is what is on the application. A health study reports by ZIP code because that is what the medical record carries. And a news story says "life expectancy varies by twenty years between these two ZIP codes" because that is the unit the underlying study used. At every step the choice is reasonable, and at no step does anyone reopen the question of what the unit is.

That is worth naming plainly, because it is the same shape as two problems this book meets everywhere. The first is the **modifiable areal unit problem** (Openshaw 1984). A statistic computed over areas depends on how the areas were cut, and ZIP areas were cut by the Postal Service for reasons having nothing to do with the statistic.

The second is the **ecological fallacy** (Robinson 1950). A relationship that holds across ZIP areas need not hold across people, and can even reverse.

A ZIP code makes both worse than usual. It is the one unit whose boundaries were drawn with no reference to any social fact at all, and the one unit whose users mostly do not know that.

12 What a routing key can testify to

The evidence here is unusually good in one specific way and narrow in several.

It is good because the crossing counts are not estimates. The relationship files are the Census Bureau's own account of how its own two geographies overlap, measured in square metres of shared land. So "this ZIP area is in three counties" is the Bureau's arithmetic, not a by-product of some tolerance chosen here. The one measurement that is the by-product of a chosen tolerance, ZIP areas against congressional districts, is reported at three thresholds so you can see how much the choice is doing.

It is narrow in six ways, and none of them is fixable with a better file.

The list of ZIP codes is a compilation, not a register. The Postal Service does not publish a free list of its own codes; the authoritative file is sold on subscription. The 41,488 codes counted here come from an open compilation assembled from address data, and it is close to right rather than right by definition. Anyone reproducing the count from a different compilation should expect a different number in the low hundreds. **That the central object of this chapter has no free authoritative list is itself one of its findings.**

Everything geographic here is the tabulation area, not the ZIP code. That is unavoidable, because the ZIP code has no shape of its own. But it means every count of crossings, splits and containments is a count about the Bureau's approximation. The Irvine comparison is the visible seam.

The crossing counts are land, not people, except where they say otherwise. Land area is what the relationship files measure, so "this ZIP area is in three counties" is a statement about ground. Where population appears it is the 2020 census count for the whole tabulation area. That answers "how many people live in a ZIP area that is split." It does not answer "how many people live on the far side of the split." That second question would need the count block by block, and it is the more interesting one.

One state's districts, in one decade. Pennsylvania under the map in force for the 119th Congress. Whether a state with fewer, larger districts splits fewer ZIP areas is not settled here and plainly could go either way.

Two cities, chosen. Irvine because the paper being replicated used it, Pittsburgh because it is where this course sits. The hundred-largest table is there so neither is taken for typical, and the hundred largest cities are not the country either.

Nothing here measures what anybody knows. The claim that voters cannot identify their own ZIP code boundary is quoted from the paper, not tested. It would take a survey, and it is a good one to want.

So the sentences this chapter will bear are these. **A ZIP code is a routing key, not a place.** Roughly one in five names no ground at all, and a fifth of the country has no ZIP code over it. The areas that do exist are the Census Bureau's guess from delivery routes. They cross counties, cross states, contain holes, arrive in pieces, and change shape between censuses without announcement.

The map they draw is a real map of the postal system, and Figure 1 is not a trick. It is not a map of anything else. Anyone who uses the unit inherits a hundred sorting decisions made by an agency whose one aim was the cost of moving mail.

Three things stay open. Whether voters recognise ZIP boundaries, which is an empirical question nobody has answered well. What to use instead when a survey needs a geography respondents can supply — because "census tract" is not an answer people have. And

whether the Postal Service should publish its own list. That is a policy question with a plain answer about where the data comes from: at present the most widely used geography in American applied research is one whose definitive definition is for sale.

The letter to 20500 arrives. That is the whole of what the five digits promise, and it is the only thing they were ever asked to do.

13 Sources

Data

- GeoNames, *Postal Codes, United States* — the list of codes in circulation. <https://download.geonames.org/export/zip/US.zip> Creative Commons Attribution. A compilation assembled from address data, not a Postal Service register; the authoritative *City State Product* is sold by USPS on subscription and is not used here.
- U.S. Census Bureau, *Gazetteer Files, 2024* — the tabulation areas, their land area and their interior points. https://www2.census.gov/geo/docs/maps-data/data/gazetteer/2024_Gazetteer/
- U.S. Census Bureau, *2020 Census Demographic and Housing Characteristics File*, table P1, at summary level 860 — the decennial count for each tabulation area, and the population behind every figure above. <https://api.census.gov/data/2020/dec/dhc> This is the one source here that needs a key. The Census data API has required one since 2025; it is free, it arrives in about a minute, and it goes in `~/ .Renvirom as CENSUS_API_KEY`. The metadata endpoints — which variables and which geographies exist — are still open, and the ZCTA row in the geography list is how the tabulation above was confirmed.
- U.S. Census Bureau, *American Community Survey 5-year estimates, 2023*, table B01003 — the second reading of the same areas, and the place populations used to rank the hundred largest cities. https://www2.census.gov/programs-surveys/acs/summary_file/2023/table-based-SF/data/5YRData/
- U.S. Census Bureau, *2020 ZCTA relationship files* — how ZCTA land divides among counties and places, and how the 2010 areas map to the 2020 ones. <https://www2.census.gov/geo/docs/maps-data/data/rel2020/zcta520/>
- U.S. Census Bureau, *Cartographic Boundary Files* — the ZCTA, place and congressional district geometry. <https://www2.census.gov/geo/tiger/GENZ2020/shp/> and <https://www2.census.gov/geo/tiger/GENZ2024/shp/>
- U.S. Census Bureau, *TIGER/Line Shapefiles, 2020, tabulation blocks* — the Pennsylvania block file, from which the blocks of Figure 2 and their population come. A 409 MB download for one county's worth of blocks; there is no smaller unit of it. <https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/>
- U.S. Census Bureau, *ZIP Code Tabulation Areas* — the delineation rules quoted in “How a ZIP code becomes a ZCTA”: the majority-of-addresses assignment, the two-square-mile threshold, and the requirement that a ZCTA code be a ZIP code valid on 1 January 2020. <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/zctas.html>

Every Census source above is keyless except the decennial counts, which come through the API. The data used for the Pittsburgh figures is the `pgh_rings.csv` and `pgh_blocks.csv` files; the digit map is drawn from `scf.csv`.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`churn.csv`, `cities.csv`, `cities_big.csv`, `digit_rows.csv`, `lists.csv`, `nesting.csv`, `no_ground.csv`, `no_ground_kind.csv`, `pa_split.csv`, `pgh_blocks.csv`, `pgh_ids.csv`, `pgh_rings.csv`, `scf.csv`, `zip_vs_zcta.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `lists.csv` counts the postal codes in circulation and the areas that have been drawn. The two do not match. Work out the gap and what falls into it.
- `nesting.csv` counts the ZCTAs that cross a county line and a state line, and the population in them. Look at `pct_pop_multi_county`. What happens to any analysis that treats a ZIP code as sitting inside one county?
- `no_ground.csv` and `no_ground_kind.csv` cover codes that name no ground at all — a single building, a post office box. Find the state with the highest share. What does a map of ZIP codes do with those?
- `pa_split.csv` counts ZIP areas split across congressional districts and the population living in them. If a campaign bought a mailing list by ZIP code, how many of those letters land in the wrong district?

Academic literature

- Curiel, J. A., and T. Steelman (2018). "Redistricting Out Representation: Democratic Harms in Splitting Zip Codes." *Election Law Journal* 17(4): 328–353.
- Curiel, J. A., and T. Steelman (2020). "A Response to 'Tests for Unconstitutional Partisan Gerrymandering in a Post-Gill World' in a Post-Rucho World." *Election Law Journal* 19(1): 101–109.
- Grofman, B., and J. Cervas (2021). "ZIP Codes as Geographic Bases of Representation." *Election Law Journal* 20(1). doi:10.1089/elj.2020.0687
- Openshaw, S. (1984). *The Modifiable Areal Unit Problem*. Norwich: Geo Books.
- Robinson, W. S. (1950). "Ecological Correlations and the Behavior of Individuals." *American Sociological Review* 15(3): 351–357.

Further reading

- Grubestic, T. H. (2008). "Zip Codes and Spatial Analysis: Problems and Prospects." *Socio-Economic Planning Sciences* 42(2): 129–149.
- Stephanopoulos, N. O. (2012). "Redistricting and the Territorial Community." *University of Pennsylvania Law Review* 160(5): 1379–1477.

Watched

- U.S. Census Bureau, "DATA GEMS: What is a ZCTA? Making Sense of Census Geography." The Bureau explaining the tabulation area in its own words, which is the distinction this chapter spends its length on. <https://youtu.be/dluu8NcTyrM>
- Matt Forrest, "Your ZIP code isn't about you, it's about a mail truck." <https://www.instagram.com/reel/DZNsovSJGpG/>

Checks

Check	Value
postal codes in the compilation	41,488
ZIP Code Tabulation Areas in the gazetteer	33,791
codes in both lists	33,642
postal codes with no tabulation area	7,846
2020 census population counted inside a tabulation area	334,726,586
median gap, ACS estimate against the census count	6%
land under no tabulation area	1,845,990 sq km, 20.2% of the country
of it in pieces over two square miles, the Bureau's stated threshold	99.99%
ZIP areas reaching into two or more counties	10,180 of 33,791
ZIP areas reaching into two or more states	137
ZIP areas that changed shape between 2010 and 2020	30,912 of 32,940
15213 rebuilt from its own census blocks	301 blocks, 31,075 people – the published count exactly
Pennsylvania ZIP areas split between districts (2% threshold)	223 of 1,833

Each line is asserted before any of the tables above are written; a change in any source that moved one of these numbers would stop the document rather than change it quietly.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE SOURCES. Two lists that are usually assumed to be one list, and the Census Bureau's own account of how they relate. Everything is open except one request, which needs a free key.

1. GeoNames, "Postal Codes, United States":

<https://download.geonames.org/export/zip/US.zip>

The Postal Service does not give away its own list of ZIP codes; it sells it. This CC-BY compilation is the list open work uses instead, and it is a compilation, not the register. One row is one postal code, tab-delimited, with a place name and a point.

2. U.S. Census Bureau, 2024 Gazetteer, ZIP Code Tabulation Areas:

https://www2.census.gov/geo/docs/maps-data/data/gazetteer/2024_Gazetteer/2024_Gaz_zcta_national.zip

One row is one ZCTA – the Bureau's built-from-blocks

approximation of a ZIP code – with its land area and a point.

3. The 2020 Census count for every ZCTA, from the API:

<https://api.census.gov/data/2020/dec/dhc>

with `get=P1_001N` and `for=zip code tabulation area:*` (summary level 860). THIS ONE NEEDS A KEY. It is free: sign up at https://api.census.gov/data/key_signup.html and append `&key=...` to the request.

4. The Bureau's 2020 relationship files, keyless, which record in square metres how ZCTAs overlap counties and how the 2010 areas became the 2020 ones:

https://www2.census.gov/geo/docs/maps-data/data/rel2020/zcta520/tab20_zcta520_county20_natl.txt

https://www2.census.gov/geo/docs/maps-data/data/rel2020/zcta520/tab20_zcta510_zcta520_natl.txt

WHAT TO BUILD.

1. The two lists side by side: how many postal codes are in

circulation, how many tabulation areas exist, how many codes sit on both lists, and how many postal codes have no tabulation area at all – real mail routes that name no ground.

2. The total 2020 population living inside any tabulation area.

3. From the relationship files: how many ZCTAs reach into more than one county, and how many into more than one state.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 41,488 postal codes against 33,791 tabulation areas, with 33,642 codes on both lists and 7,846 postal codes naming no ground
- 334,726,586 people counted inside a tabulation area
- 10,180 ZCTAs reach into two or more counties, and 137 into two or more states

The GeoNames compilation moves week to week, so the postal-code side will drift with the day you fetch it; the census side is frozen until the 2030 ZCTAs are drawn. Drift on the first list is the list changing, not a mistake.